Navigating Airflow's user interface

In this train, we'll learn how to navigate Airflow's UI.

Learning objective

> - Understand how to navigate Airflow's UI.

Accessing the Airflow UI

Apache Airflow provides a web-based user interface (UI) that allows users to manage and monitor workflows efficiently. Steps:

1. Have a running Airflow server or start one by running `airflow webserver -p 8080`
2. Open your browser and go to: `http://localhost:8080`
3. Log in using your credentials if authentication is enabled.

Understanding the Airflow Dashboard

Once logged in, you will see the main dashboard, which consists of several sections:

- **DAGs View:** Lists all available Directed Acyclic Graphs (DAGs).
- **Status Indicators:** Shows the health of DAG runs (Success, Running, Failed, etc.).
- **Last Run and Next Run:** Provides scheduling details for each DAG.
- **Actions Column:** Includes options to trigger, pause, and delete DAGs.

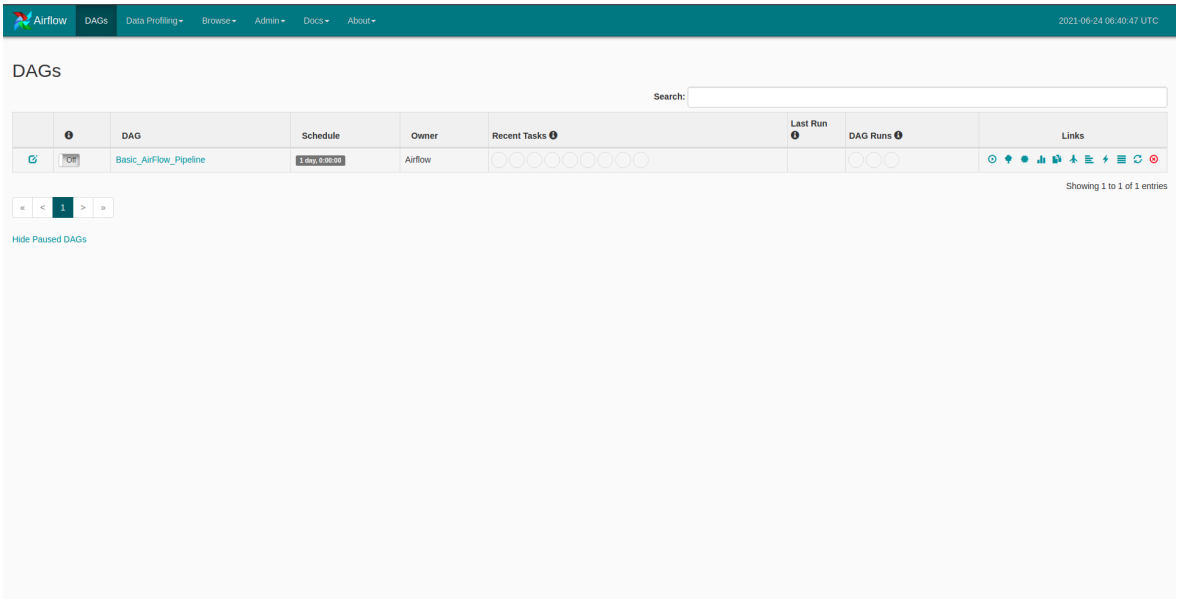


Figure 1: The Airflow UI's overview page.

Running DAGs

To run your pipeline/DAG there are two quick actions you need to perform on the Airflow landing page. As illustrated in Figure 2, you initially need to enable your DAG by selecting the on/off toggle switch to the left of the screen corresponding to your DAG's name ("Basic_Airflow_Pipeline"). Following this, you can trigger the pipeline by selecting the small play button on the right of the screen.

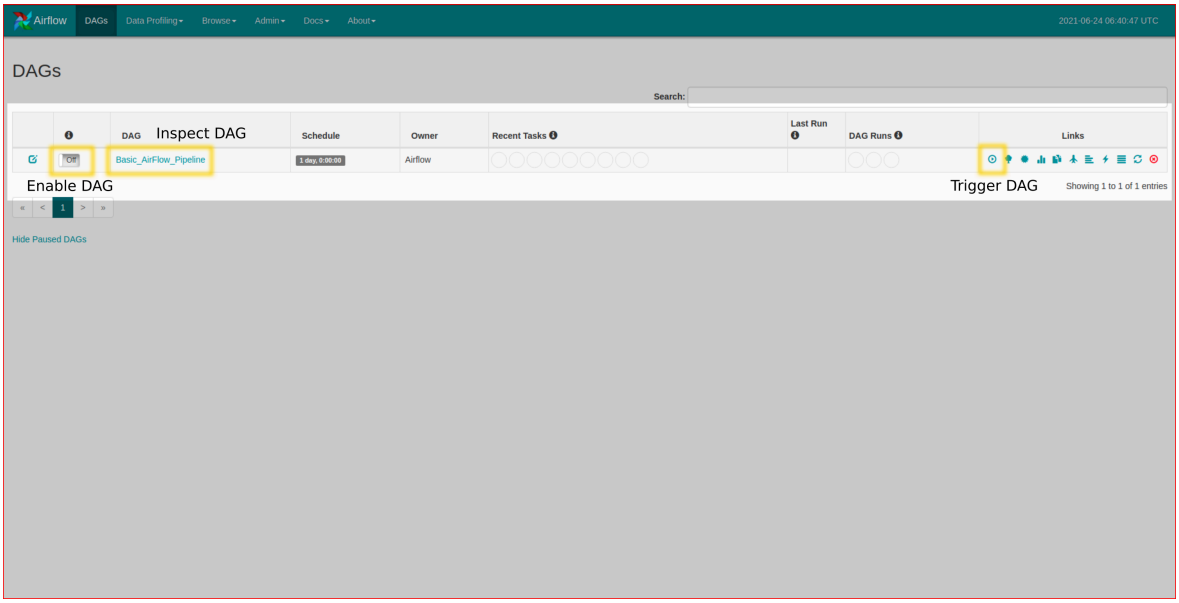


Figure 2: The Airflow UI overview page with triggering mechanisms annotated.

Once triggered, Airflow will begin humming away in the background, and you should see the terminal output indicating various operations being performed. After a couple of seconds, hit the refresh button associated with the DAG (near the trigger button), and you should see the UI indicating that the task has executed successfully, as represented in Figure 3.

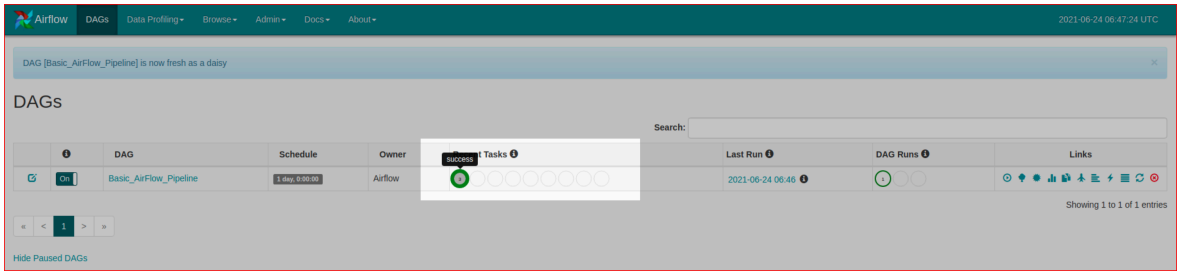


Figure 3: The Airflow UI overview page with all tasks having successfully completed.

Monitoring DAGs

One of the powerful features of Airflow is its ability to present immense detail surrounding the performance and state of DAGs through the Airflow UI. Having successfully run your workflow, let's take a quick tour around the Airflow UI to get further insight into how your pipeline executed.

The DAG tree view

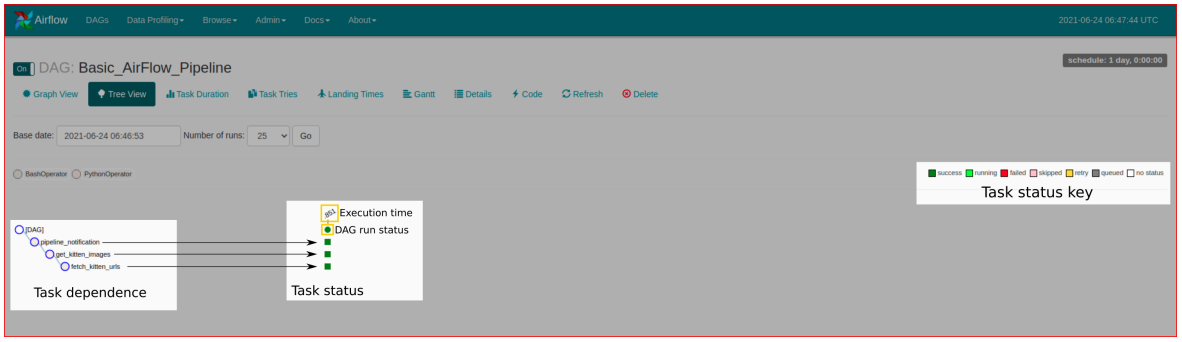


Figure 4: The tree view of the successfully completed DAG. Annotations are provided to help show the representational power of this visualisation.

The first visual you'll inspect is the tree view. You can navigate to this page by clicking on your DAG's name from the landing screen, or by selecting the Tree View tab when already inspecting a DAG.
The tree view provides two interrelated visuals.
On the left, a hierarchical representation of each task's dependence within the DAG is given, with each branch being executed from the leaf node (bottom of the branch) to the root node (top of the branch). The presence of multiple branches (not shown in Figure 12) represents how tasks within your DAG can run independently of one another until depending on a common task node.
On the right of the tree view, the second visual provides a growing grid-like display that increases each time your DAG is run. Here, each row of the grid represents a single task within your DAG (aligned to the tree visual). Each column of the grid represents a point in time at which the entire DAG was run. Combined, each cell within the grid represents the status of a given task at a point in time in which the entire DAG was run. This gives a powerful view of how your pipeline is performing over time and where problems may be arising due to specific task failures.

The DAG graph view

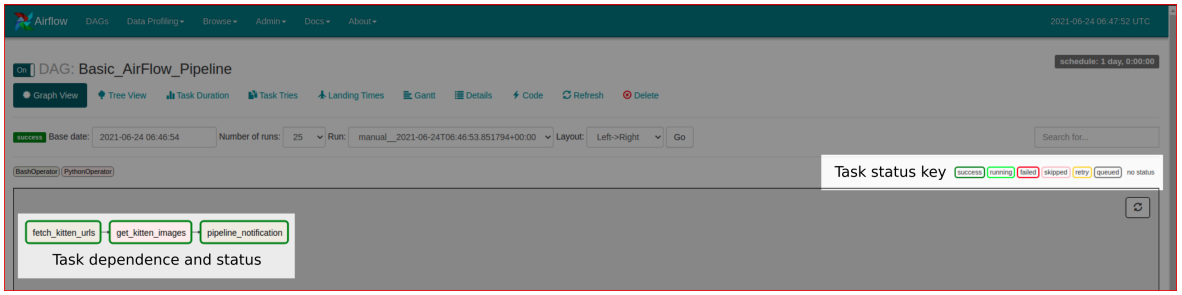


Figure 5: The graph view of the successfully completed DAG, as visualised within the Airflow UI.

By clicking on the Graph View tab, you can explore the next visual in the Airflow UI. Whereas the tree view helps you visualise the state of various tasks over time, the graph view provides a full representation of your DAG during its current run cycle. This lets you visually map out the status of various tasks and their downstream dependencies.

Viewing your notification task output

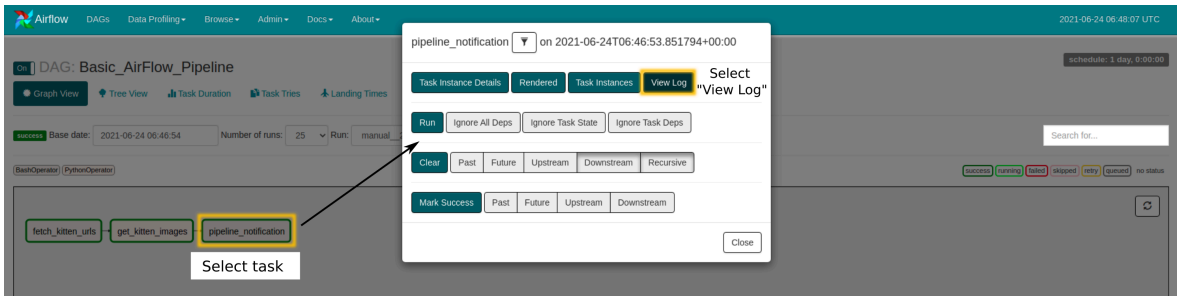


Figure 6: Navigating to a task's log data within the graph view of the Airflow UI.

While in the graph or tree views of the Airflow UI, you can also retrieve detailed information around a given task by inspecting its logs. Let's see an example of this by checking that your pipeline_notification task is indeed correctly reporting the number of kitten images you've downloaded. To do this, click on the pipeline_notification task from the graph view (or the corresponding cell within the tree view), and then select "View log". This process is represented in Figure 6.

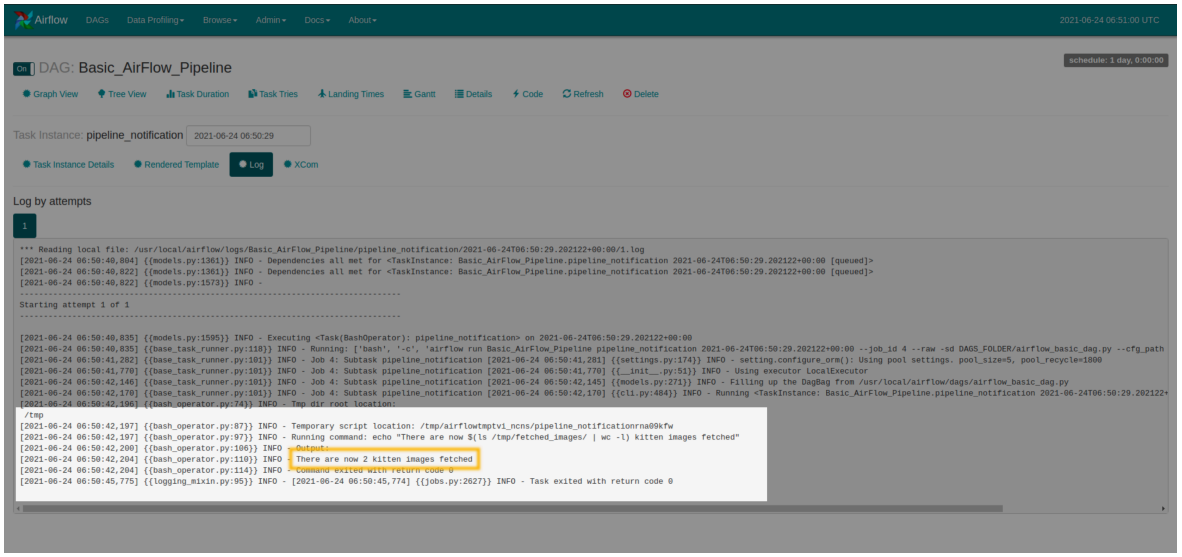


Figure 7: Viewing the log-based output of a 'pipeline notification' task within the Airflow UI.

Once you've chosen to view the task log, you should see a large amount of verbose text representing the logging information captured by Airflow. As shown in Figure 7, scrolling towards the bottom of this log should reveal the line echoed by your notification task, stating that there are a certain number of images fetched (this should be the number of times the pipeline has been run multiplied by two).

Additional views

Besides the graph and tree views that you've covered, there are several other visuals which Airflow provides to help you monitor, debug, optimize, and report on your workflow. We encourage you to use your environment as a sandbox and to explore these additional visuals under the tabs, such as "Task Duration", Task Tries, Landing Times, and Gantt.

Conclusion

By following these steps, you can effectively navigate and manage workflows using the Apache Airflow UI. Mastering this interface will enhance your ability to monitor, troubleshoot, and optimize data pipelines.

Key Takeaways

- **Understanding the dashboard:** The Airflow UI provides a clear overview of all DAGs, their statuses, and scheduling details, making it easy to monitor workflows.
- **Running and monitoring DAGs:** Users can trigger, pause, and inspect DAGs through views like Graph View and Tree View, ensuring efficient debugging and performance tracking.
- **Task logs and debugging:** Airflow's logging feature allows users to drill down into task execution details, helping diagnose issues and optimize workflow performance.